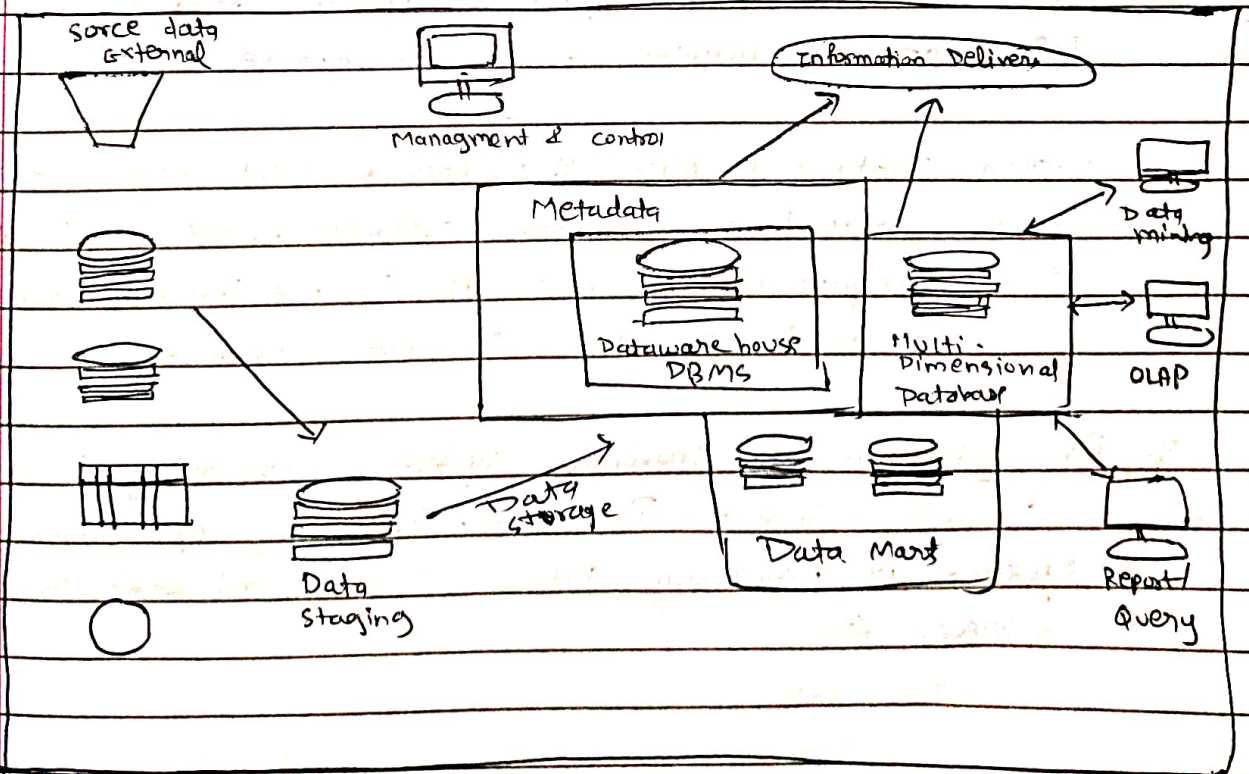
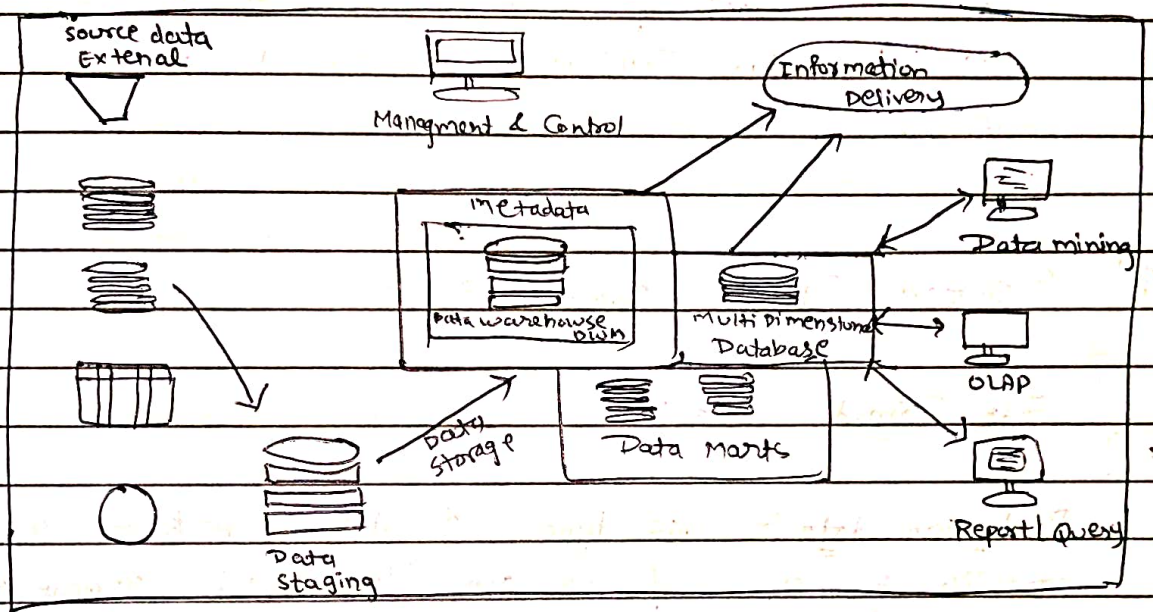


Theory DWM

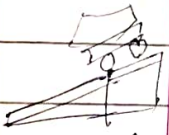
Q1. What are Basic Building Blocks of Data warehouse?

Ans. Architecture is the proper arrangement of elements. We build a data warehouse with software and Hardware components.



The fig shows essential elements of a Typical warehouse. We see the Source Data Component on left. The Data staging element serves as the next Building block. In the middle, we see the Data Storage Components that Handles data warehouse data. This elements not only storage and manage the data; it also keep Track of data using metadata ~~Res~~ Repository. The information Delivery Component shows on the Right.

(D) Source Data Component :-



Four Broad Categories:

1. Production data :- This type of data come from the Different O.S of enterprise. You select segment of data from various Operational system based on Information Requirement in Data warehouse.
2. Internal Data :- In each organization, client keeps their "private" spreadsheet, Reports, customer profile, and sometime even Department database. This is the internal data which could be useful in data warehouse.
3. Archived Data :- Operational System are mainly intended to run current Business. In every operational system, we periodically take old data and store in achieved files.

4. External data $\div$ Most executives depends on information from External ~~data~~ source of a large percentage of information they use. They rely on competitor market share data.

(II) Data Staging Component

After we have been extracted data from various operational system and external sources, we have to prepare files for storing in the data warehouse.

We will now discuss three primary function takes place in staging area.

(a) Data Extraction

(b) Data Transformation

(c) Data Loading

(1) Data Extraction $\div$ This method had to deal with numerous data sources. We have to employ the appropriate technique for each data source.

(2) Data Transformation $\div$ Data Transformation in a data warehouse involve several task, including cleaning data to correct errors, handle missing values, Remove duplicates. Sorting, merging and Restructuring data are ~~comp~~ common steps in the staging area.

The goal is to produce integrated, cleaned, standardized and summarized data Ready for analysis.

In data dictⁿ we keep data about logic & data size

In table 'd' in data of logical dⁿ



(3) Data loadings = Two distinct categorize of Tasks form data loading functions. When we complete the structure and construction of data warehouse and go live for first time, we do initial loading of Information into Dataware storage. The initial ~~data~~ loads moves High volume of data using up a ~~subsequent~~ substantial amount of time.

(III)

(a) Data Storage Component

Data storage of data warehousing is a split Repository. The data Repositories for the operational system generally include only the current data. Also these Data Repositories includes data structured in Highly normalized for fast and efficient processing.

(b) Metadata Component

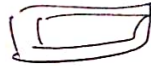
Metadata in a data warehouse is equal to Data dictionary ~~and~~ or data catalog in database Management system. In data dictionary, we keep data about the logical data structure.

② Data Marts

- It is a sub-set of corporate-wide data tailored for specific user groups
- Focused on selected subjects, smaller than data warehouse and support specific queries and Reports.

(a) ~~Data~~ Management & Control Component

- Coordinates data Transformation, Transfer and Delivery
- Ensure data is properly stored and monitors its movement from staging area of the ^{dm} Warehouse.



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Q. Explain Page Ranking Technique in Detail

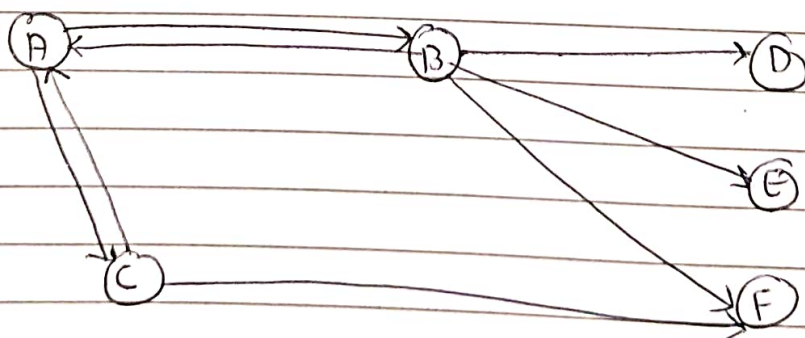
- Ans
- (i) The page Ranking algorithm is applicable to web pages.
 - (ii) The page Rank algorithm is used by Google search to rank many website in their search engine results.
 - (iii) The page rank algorithm was named after Larry Page, one of the founder of Google.
 - (iv) We can say that the page rank algorithm is a way of measuring the importance of website pages.
 - (v) A web page Basically is a directed graph which has two components nodes and connections.
[pages $\Rightarrow$ nodes
Hyperlinks $\Rightarrow$ connections]

Algorithm

- We assume page A has pages B...N which point to it.
- The parameter d is damping factor / Teleportation factor which can be set b/w 0 and 1. We usually set it 0.85.
- Also, $\text{Count}(A)$ is Defined as number of links going out of page A.
- A page with not link out is ~~called~~ called as Dead end.
- A Spider Trap is a set of ~~and~~ node with no dead end but no arcs out. Spider Traps make an infinite number of Request or cause a poorly constructed crawls to crash.
- Both Dead end and Spider Traps are avoided.
- The Page Rank of Page A is as follows:

$$\text{PR}(A) = (1-d) + d \left(\text{PR}(B) / \text{Count}(B) + \text{PR}(C) / \text{Count}(C) + \dots + \text{PR}(N) / \text{Count}(N) \right)$$

Eg: Compute Page Rank at every node at end of second iteration. Use Teleportation factor $\Rightarrow 0.8$



using the formula: $PR(A) = (1-d) + d (PR(B)/\text{Count}(B) + PR(C)/\text{Count}(C) + \dots + PR(N)/\text{Count}(N))$

$$\therefore PR(A) = (1-0.8) + 0.8 * (PR(B)/4 + PR(C)/2)$$

$$PR(B) = (1-0.8) + 0.8 * (PR(A)/2)$$

$$PR(C) = (1-0.8) + 0.8 * (PR(A)/2)$$

$$PR(D) = (1-0.8) + 0.8 * (PR(B)/4)$$

$$PR(E) = (1-0.8) + 0.8 * (PR(B)/4)$$

$$PR(F) = (1-0.8) + 0.8 * (PR(B)/4 + PR(C)/2)$$

iteration 0

Assume that each page having page rank $\Rightarrow 1/\text{Total no of nodes}$

$$\text{Therefore } PR(A) = PR(B) = PR(C) = PR(D) = PR(E) = PR(F) = 1/6 = 0.167$$

Iteration 1

$$PR(A) = (1-0.8) + 0.8 * (0.167/4 + 0.167/2) \Rightarrow 0.3$$

Now we use updated Page Rank for further calculation

$$PR(B) = (1-0.8) + 0.8 * (0.3/2) \Rightarrow 0.32$$

$$PR(C) = (1-0.8) + 0.8 * (0.3/4) \Rightarrow 0.3$$

$$PR(D) = (1-0.8) + 0.8 * (0.3/4) \Rightarrow 0.264$$

$$PR(E) = (1-0.8) + 0.8 * (0.3/4) \Rightarrow 0.264$$

$$PR(F) = (1-0.8) + 0.8 * (0.3/4 + 0.3/2) \Rightarrow 0.392$$

$$PR(A) \Rightarrow 0.3$$

$$PR(B) \Rightarrow 0.32$$



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Iteration 2

$$PR(A) = (1 - 0.8) + 0.8 * (0.32/4 + 0.32/2) \Rightarrow 0.392$$
~~$$PR(B)$$~~

Now use the update Page Rank for further calculation

$$PR(B) = (1 - 0.8) + 0.8 * (0.392/2) \Rightarrow 0.3568$$

$$PR(C) = (1 - 0.8) + 0.8 * (0.392/2) \Rightarrow 0.3568$$

$$PR(D) = (1 - 0.8) + 0.8 * (0.3568/4) \Rightarrow 0.2714$$

$$PR(E) = (1 - 0.8) + 0.8 * (0.3568/4) \Rightarrow 0.2714$$

$$PR(F) = (1 - 0.8) + 0.8 * (0.3568/4 + 0.3568/2) \Rightarrow 0.4141$$

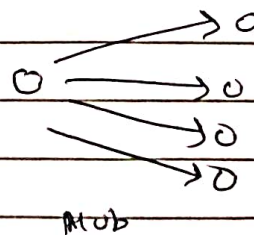
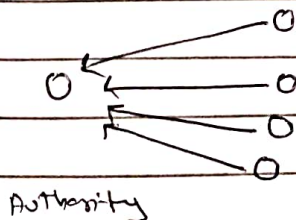
Nodes	Iteration 0	iteration 1	iteration 2
A	1/6	0.3	0.392
B	1/6	0.32	0.3568
C	1/6	0.32	0.3568
D	1/6	0.264	0.2714
E	1/6	0.264	0.2714
F	1/6	0.392	0.4141

(II) HITS Algorithm

- (i) HITS stands for Hyperlink-Induced Topic Search.
- (ii) ~~HITS~~ HITS, also known as Hubs and authorities, developed by Jon Kleinberg is a link analysis algorithm that rates web pages. It was a precursor of Page Rank.
- (iii) The idea of this algorithm originated from the ~~use~~ fact that an ideal website should be linked to other relevant web sites and also being linked by other important sites.
- (iv) HITS uses Hubs and Authority to Recursive Relationship b/w webpage.

Authority: A node is High Quality if many High Quality nodes link to it.

Hubs: A node is High Quality, if it links to many High-Quality nodes.



$$H = a^T H$$

$$A = a^T H$$

$$H \Rightarrow a^T A$$

$$A \Rightarrow a^T H$$

$$H = a^T A$$

$$A = a^T A H$$

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Eg: Adjacent matrix for a Graph of four vertices (n=4) is given below:

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Calculate authority & Hub scores for this graph using HITS Algorithm with $k=6$ & Identify the best authority & hub nodes

$$Adj = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$Adj^T = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

For $k=1$

$$h = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$Authority\ vector = a^T h = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 4 \end{bmatrix}$$

$$H = a^T A$$

$$A = a^T H$$

$$A = \begin{bmatrix} 1 \\ 1 \\ 2 \\ 4 \end{bmatrix}$$

The updated hub vector $\div a^T A$

$$h = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 7 \\ 6 \\ 5 \\ 4 \end{bmatrix}$$

For $\lambda = 2$;

Authenticity vector = $\text{adj}^T \times A h$

$$\Rightarrow \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \times \begin{bmatrix} 7 \\ 6 \\ 5 \\ 4 \end{bmatrix} \Rightarrow \begin{bmatrix} 25 \\ 7 \\ 13 \\ 22 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 & 0 & 7 & 0 \\ 6 & 0 & 0 & 0 \\ 13 & & & \\ & & & \end{bmatrix}$$

Hub = $a \times A$

$$= \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 7 \\ 13 \\ 22 \end{bmatrix} = \begin{bmatrix} 42 \\ 35 \\ 27 \\ 22 \end{bmatrix}$$

For $k=3$

Adjacent vector = $a^T H$

$$= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 42 \\ 35 \\ 27 \\ 22 \end{bmatrix} \Rightarrow \begin{bmatrix} 27 \\ 42 \\ 77 \\ 126 \end{bmatrix}$$

Hub vector = $a * A$

$$= \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 27 \\ 42 \\ 77 \\ 126 \end{bmatrix} \Rightarrow \begin{bmatrix} 245 \\ 203 \\ 153 \\ 126 \end{bmatrix}$$

For $k=4$

Adjacent vector = $a^T H$

$$= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 245 \\ 203 \\ 153 \\ 126 \end{bmatrix} = \begin{bmatrix} 153 \\ 245 \\ 448 \\ 727 \end{bmatrix}$$

Hub = $a * A$

$$= \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 153 \\ 245 \\ 448 \\ 727 \end{bmatrix} = \begin{bmatrix} 1420 \\ 1175 \\ 880 \\ 727 \end{bmatrix}$$

K=5

$$\text{Authority} = a^T H$$

$$= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1420 \\ 1175 \\ 980 \\ 727 \end{bmatrix} = \begin{bmatrix} 980 \\ 1420 \\ 2595 \\ 4202 \end{bmatrix}$$

$$\text{Hub} = a * A$$

$$= \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 980 \\ 1420 \\ 2595 \\ 4202 \end{bmatrix} = \begin{bmatrix} 8217 \\ 6797 \\ 5082 \\ 4202 \end{bmatrix}$$

K=6

$$\text{Adjacency vector} = a^T * H$$

$$= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 8217 \\ 6797 \\ 5082 \\ 4202 \end{bmatrix} = \begin{bmatrix} 5082 \\ 8217 \\ 15014 \\ 24298 \end{bmatrix}$$

$$\text{Hub} = a * A$$

$$= \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5082 \\ 8217 \\ 15014 \\ 24298 \end{bmatrix} \Rightarrow \begin{bmatrix} 47529 \\ 39312 \\ 29386 \\ 24298 \end{bmatrix}$$

Hence, after 6 iteration, we can say that the best authority node is n_4 (24298) and best hub n_1 (47529) having Highest value

Q Compare OLTP and OLAP

Ans

Category	OLAP	OLTP
1. Definition	1. It is well known as an online database Query management System	1. It is well known as an online database modifying system
2. Data Source	2. Consist of Historical data from various databases.	2. Consist of only operational current data
3. Normalized	3. In OLAP, database, Table is not normalized	3. In OLTP database, table are normalized
4. Operation	4. Only Read and rarely write operations	4. Both read and write operation
5. Volume of data	5. large amount of data is stored typically in TB, PB	5. Small amount of data is store typically in MB and GB.
6. Database Type	6. It uses data warehouse	6. It use Traditional DBMS
7. Response Time	7. Processing is little slow	7. In Milliseconds
8. Query Type	8. Complex Queries	8. Simple Queries
9. Concurrency	9. Support Thousands ^{Few} of Concurrent user	9. Support fewer ^{Thousand of} Concurrent user.
10. operation	10. SELECT Command	10. Based on INSERT, UPDATE, DELETE Commands

Q Differentiate H/w Agglomerative and Divisive clustering method.

Aspects	Agglomerative cluster	Divisive cluster
1. Approach	1. Bottom - up approach	1. Top-down approach
2. Initial step	2. Each data point starts as its own cluster.	2. All Data points are grouped into one cluster
3. Tree Construct	3. Build Hierarchical Tree upward	3. Build Hierarchical Tree Downward
4. Complexity	4. Less computationally complex	4. More computationally complex.
5. Visualization	5. Dendrogram starts with as many leaves and end at the Root	5. Dendrogram starts with Root and end with many leaves
6. clustering speed	6. Faster	6. Slow
7. Noise Handling	7. More sensitive to noise	7. can Handle the Noise Better.
8. Clusters stability	8. More stable Result	8. less stable results
9. Space handling	9. works well in ^{low} Dimension	9. works better in High-Dimension.
10. Memory usage	10. less memory Intensive	10. More Memory Intensive

Q Discuss Data visualization Technique

- Ans
- (i) Visualization is the conversion of data into visual or Tabular format
 - (ii) Visualization of data is one of the most powerful and appealing Technique for data exploration
 - (iii) Data visualization can identify area that need improvement or modification.
 - (iv) Data visualization can predict sales volumes.

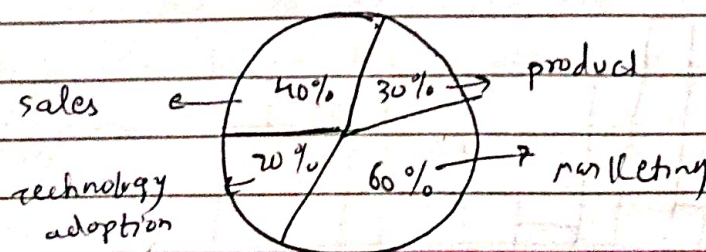
Techniques in visualization

- pie chart
- Bar chart
- Histogram
- Gantt chart
- Scatter plot
- Pictogram chart
- Star plot
- Box plot

(1) Pie chart

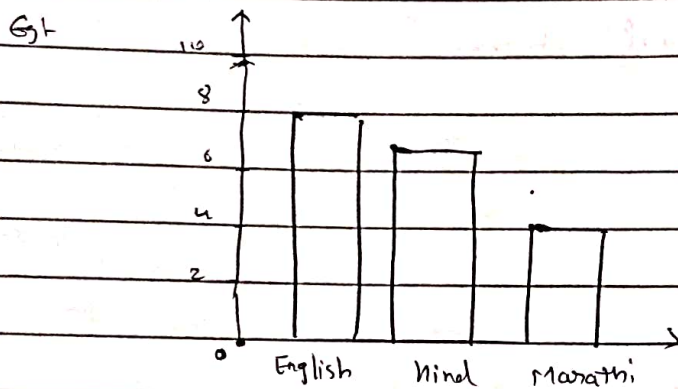
- A pie chart is a circular statistical graphic Representation, and it is very simple and easy to understand.

For eg: A Company witness a growth of 150% in which they found out 60% growth due to marketing, 40% Growth due to sales, 30% Growth due to product and 20% was due to Technology adoption.



(2) Bar chart

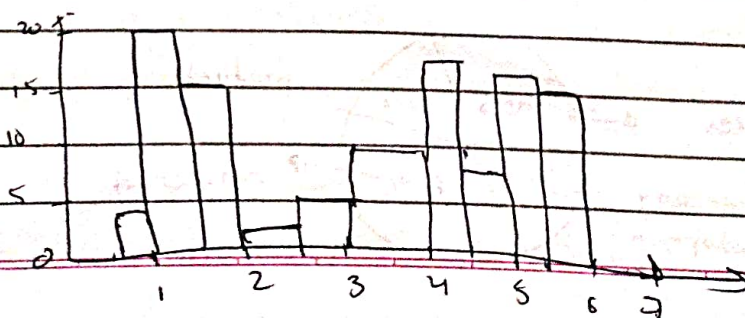
- (i) It is one of the simple Technique of data visualization.
- (ii) These chart are used to compare the Quantities of Different Categories
- (iii) The value of categories are addressed with the bars and they can be designed with vertical or flat bars, with length ~~error~~ Height addressing the value



(3) Histogram

- (i) Graphical Representation of information using bars of various Height is Histogram
- (ii) Taller bar indicate more data fall in that Range
- (iii) A Histogram Display numericals data by grouping data into "bin" of equal width.
- (iv) Shapes of Histogram Depends on the number of bins.

Eg: Petal width and petal length

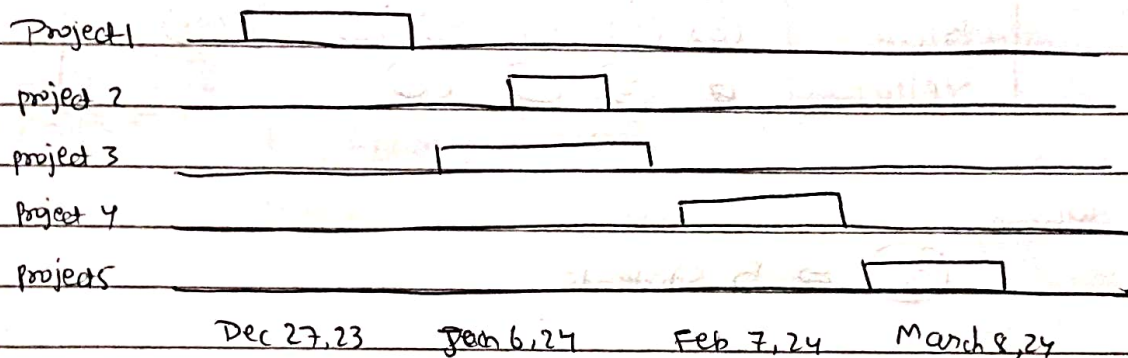


4) Gantt chart

- (i) A Gantt chart is a Horizontal bar chart that visually Represent a project plan over time.
- (ii) It keeps a track of status of each Task in project, and who is Responsible for which task, which helps teams Deliver Project on time.
- (iii) Gantt chart illustrate the start and end dates of the element and Summary element of project.

(iv)

G1-



5) Pictogram chart

- (i) The pictograph is a method of Represent the data using image.
- (ii) Pictograph Represent the data using image and is extremely ^{easy} ~~simple~~ to understand, and it is simple way to Represent statistical data.

Eg:- It will tell us how to interpret data using pictograph, In fig, the data of 100 students is collected, who like different colors.

No. of students	color
20	Red
30	Black
35	Blue
15	Yellow

The Pictograph Representation

color	no. of student
Red	   
Black	     
Blue	      
Yellow	   

Where,

 $\Rightarrow$ 5 students.

(i) Box plot

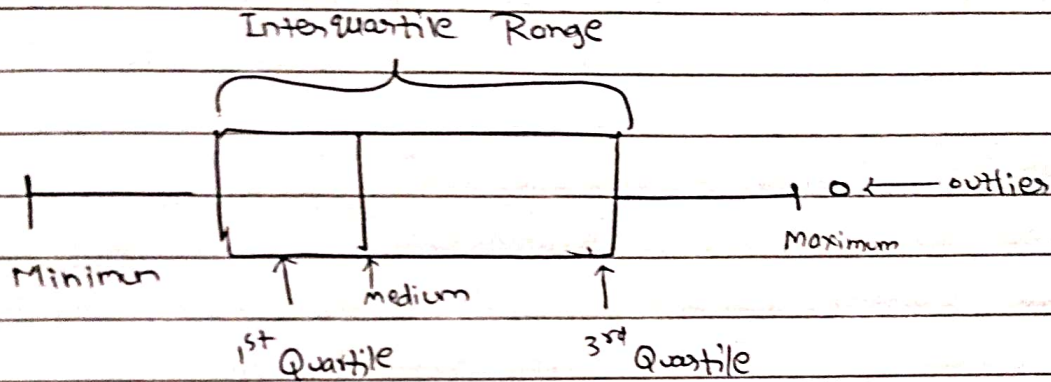
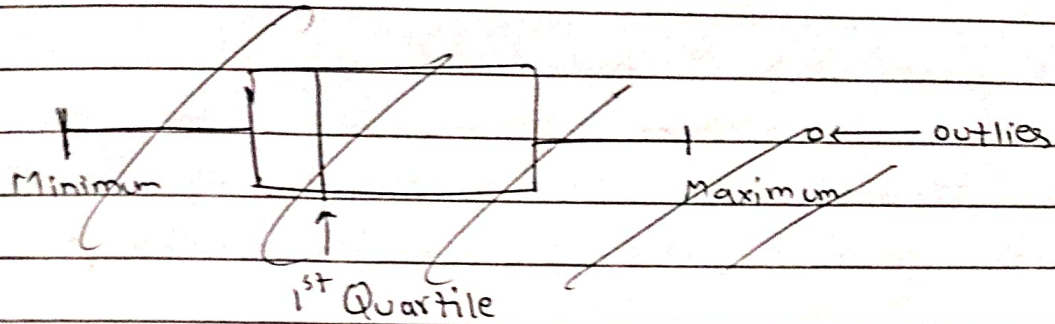
(i) This plot can be used to obtain more statistical details about the data.

(ii) The straight line at maximum and minimum are also called whiskers.

(iii) Points that lies outside the whiskers will be consider as outliers.

(iv) With the Help of Box plot, we can also determine the Interquartile Range (IQR) where maximum details of data will be Present.

∴ It will give clear idea about the outliers in dataset



DATA QUALITY
Data entry errors, Data storage problems, Data transmission problems



Q. Issue in Data mining?

Ans (1) Data Quality

- (i) The Quality of data used in data mining is one of the most significant challenges.
- (ii) Data Quality issues can arise due to variety of Reasons, including data entry errors, data storage issue, Data integration problem, data Transmission errors.
- (iii) Data must be accurate, complete and consistent for genuine data mining measure

Solution

- Implement Robust Data cleaning method to preprocess Data and improve Quality.
- Continuously monitor and update data Quality Standards

(2) Data privacy & security

- As Data mining involves analyze sensitive information, ensuring data privacy and security is crucial.
- As more data is collected, stored and analyzed, the Risk of data Breaches and cyber attacks increases

Solution:

- Encryption: Convert Data into coded format so that only authorized user with correct ~~Dec~~ Decryption key can ~~Read~~ Read it

6) Diverse Data Type Issues:

- Data has many faces, you can find it in visual, audio, text and numeric form. processing these different type and then mining may be difficult.

(4) Algorithm

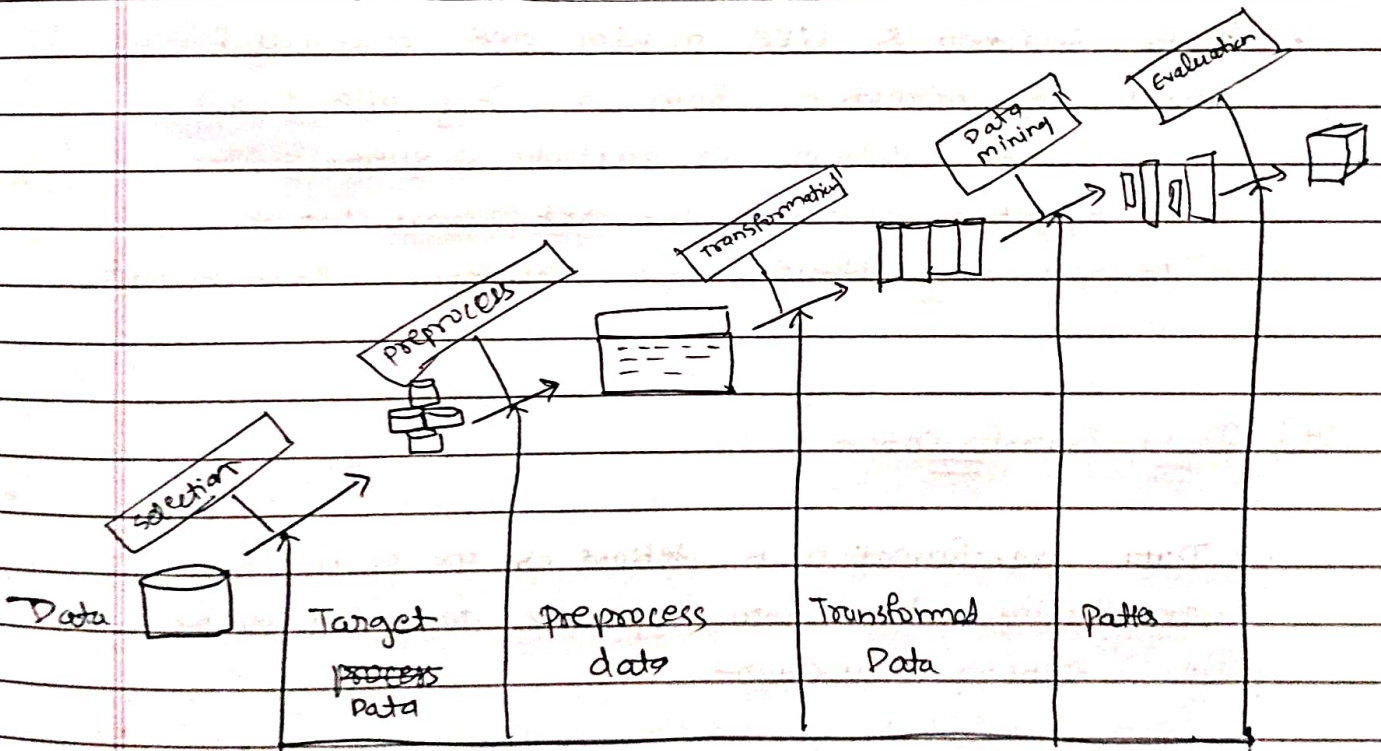
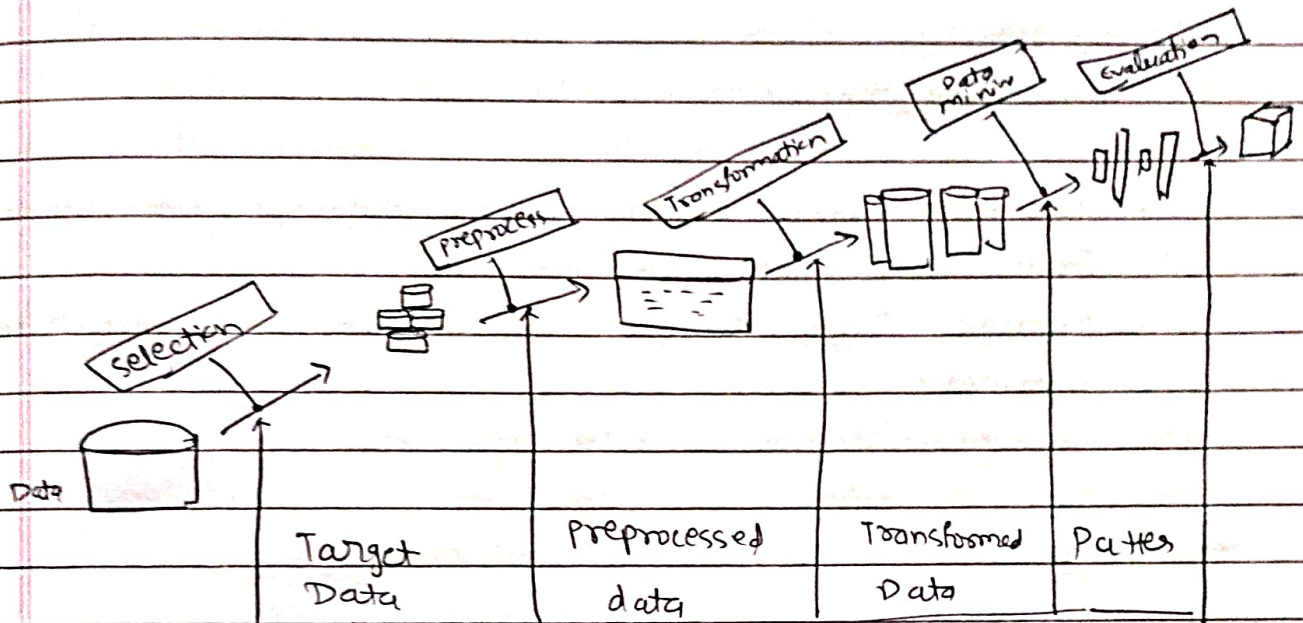
- Choosing the Right Algorithm for a specific Data mining task is not straight forward, Different algorithm have varying strengths and weakness, and selecting the wrong one can lead to suboptimal results.

(5) Dynamic Data

Data is often not static and changes over time. Mining Dynamic data requires Techniques that can adapt new information and update Model accordingly.

Q Describe the steps involved in Data mining when viewed as a process of Knowledge Discovery

Ans.



① Data cleaning :

- Remove noise, inconsistent data and outliers,
- Strategies to handle missing data fields

② Data Integration

- Data from various source such as database, datawarehouse, Transactional data are integrated.
- Multiple Data source are combined in single Data format.

③ Data selection

- Data selection is like picking the most useful piece of information from a Big pile of data to solve problem or answer a question.
- For selection we can use ~~NET~~ Neural Network, Decision Tree, Naive Bayes, clustering, Regression method

④ Data Transformation :-

- Data Transformation is defined as the process of transforming data into appropriate form required by Data mining procedure

⑥ Data mining

- This is the Heart of KDD process involves applying various Data mining Techniques to the Transformed data to Discover Hidden patterns, Relationships & Insights.
- A few of the most common data mining Technique include clustering, classification, association rule, anomaly Detection.

⑥ Data Evaluation or Pattern Evaluation

- Not all pattern are useful or interesting. This step Helps filter out the noise, ensuring only valuable insight are considered.

⑦ Knowledge Presentation

After all the Hard work, it's time to share the Findings. This often involves visualization, reports or other means to make Knowledge accessible and understandable.

DATE: / /

Q. Differentiate b/w star schema and snowflake schema

Star schema	snowflake schema
1. In star schema, The fact Table and Dimension Table are contained.	1. In snowflake schema, The fact Table, Dimension Table as well as sub-Dimension Table are contained.
2. Star schema is Top-down model	2. Snowflake schema is Bottom-up approach.
3. Uses more space	3. Uses less space.
4. less time of execution of Queries	4. More time than star schema for execution of Query
5. Normalization is not used	5. Normalization & Denormalization Both are used.
6. Design is very simple	6. Design is complex.
7. Understanding is very simple	7. Understanding is Difficult
8. less Number of Foreign keys	8. More number of Foreign keys
9. Simple DB Design.	9. Complex DB Design.
10. Easier to Maintain & update	10. Challenging to maintain and update

Q. Explain Data pre-processing

- Ans (i) Data preprocessing is a data mining technique that involves transforming raw data into an understandable format.
- (ii) Real-world data is often incomplete, inconsistent, lacking in certain behaviours or Trends and likely to contain many errors.
- (iii) Data preprocessing is a proven method to solve such issues. Data preprocessing prepares raw data into further processing.
- (iv) Data goes to Series of steps during preprocessing:

Some common steps in Data preprocessing include:

- (1) Data Cleaning: This involves identifying and correcting errors or inconsistencies in the data, such as missing values, outliers and duplicates. Various Techniques are used for data cleaning such as imputation, removal or transformation.
- (2) Data Integration: This involves combining data from multiple source to make unified dataset. Data integration can be challenging as it Requires Handling data with Different formats, structures and semantics. Techniques like Record linkage and data fusion can be used for Data integration.

3. Data Transformation:- These involves converting the data into suitable format for analysis. Common Technique used in Data Transformation are normalization, standardization and Discretization.

4. Data Reduction:- This involves Reducing the size of the dataset while preserving the important information.

There are various Method for Reduce data is

(a) Encoding Mechanism, which can be used to Reduce the size of data as well. If ~~the~~ all original data can be recovered after Compression, then operation is ~~labb~~ labelled as lossless.

If some data is lost, then it is called as lossy Reduction.

(b) Aggregation can also be used, for eg:- To Condense countless Transaction into a single weekly or monthly value, significantly reducing the ^{no's} data objects.

(5) Data Discretization:- It involves Dividing continuous data into Discrete categories or intervals.

Discretization is often used in Data mining and ML algorithm.

Discretization can also be achieved ~~by~~ by using techniques such as equal width ~~labb~~ binning, (EWB) ~~equal width binning~~

CEFB Equal Frequency binning, ~~and~~ and clustering. "

it involves
dividing

Data $\rightarrow$ so for a
normal, stand & D₀ ped $\rightarrow$ A
norm, s, .D EWB, EFB & Z₀ $\rightarrow$ EWB

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dividing com
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(6) Data normalization :- This involves scaling data to common range, such as b/w 0 to 1 or -1 to 1.

Normalization is often used to handle data with different units and scales.

Common techniques includes min-max normalization, Z-score normalization, decimal scaling.

Data preprocessing plays a crucial role in ensuring the quality of data and accuracy of analysis results.

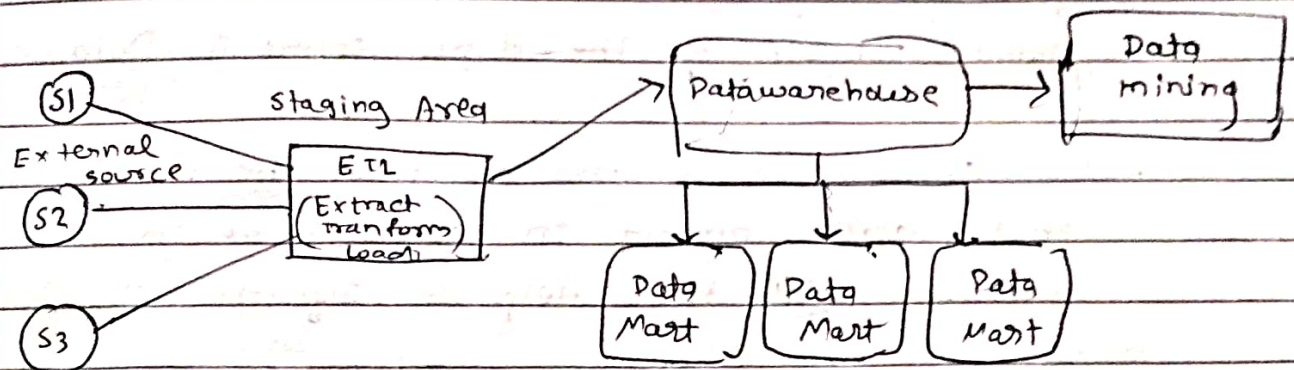
Q Difference b/w Top-down approach and Bottom-up approach for Building data ware. Discuss merits and limitation of each approach and also explain practical approach for Design Data warehouse.

(B)

Ans (A)	Top-Down Approach	Bottom-up Approach
1.	In, these problems are broken down in small parts	1. In these, the smaller problems are solved.
2.	It generally uses structured programming language such as C, COBOL, FORTRAN, etc	2. It generally used OOP such as Java, C++, Python, etc
3.	Communication among modules are less	3. Communication among modules are more more.
4.	In Top-down approach Decomposition Takes place.	4. In Bottom-up approach Composition Take place
5.	It has Higher upfront Cost.	5. It has lower initial Cost.
6.	It is More complex	6. It is less complex
7.	Focus of global view of organization.	7. Focus on local view of organization.
8.	It is less Flexible	8. It is more Flexible
9.	Easier to Maintain	9. Maintenance is Harder
10.		10.

(c) Explain practical approach for Design Data warehouse?

Ans (I) Top-Down Approach



1. External source :- External source is a source from where data is collected irrespective of the Type of data.

Data can be Structured, semi-structured, and unstructured as well.

2. Staging Area :- Since the data, extracted from external source does not follow any particular format, so there is a need to validate this data to load into datawarehouse house.

For this purpose, it is recommended to use ETL tool.

- E (Extracted): Data is Extracted from External Data source
- T (Transform): Data is Transform into Standard format.
- L (Load): Data is Loaded into datawarehouse after Transforming it into standard format.

3. Datawarehouse house :- After cleansing of data, it is store in Data-warehouse as central Repository. It actually store meta data and the actual data is stored in Data marts. NOTE that Datawarehouse stores the Data in purest form in Top-down approach

Ques can say that, data mart contain a sub-set of data which stored in Data warehouse

4) Data Mart: Data mart is also part of storage components. It stores information of a particular function of an organization which is handled by single authority. We can also say that data mart contains sub-set of the data stored in Data warehouse.

5) Data Mining: It is used to find the hidden pattern that are present in database or in data-warehouse with the help of algorithm of Data mining.

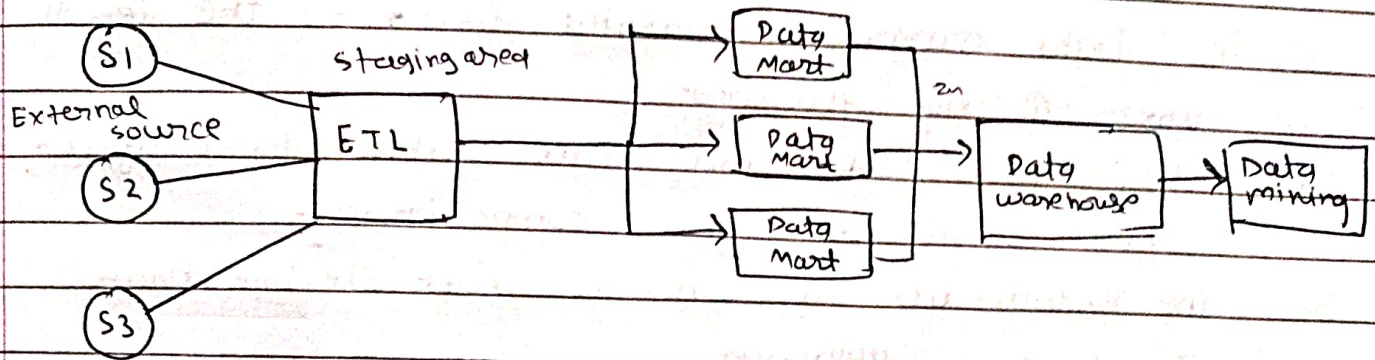
Advantages

1. Since data mart are created from data warehouse, provides consistent view of data Mart.
2. This Model is considered as strongest Model for Business changes.
3. Creating Data Mart from Data warehouse is easy.
4. Since all Data Mart are sourced from central Datawarehouse, it is easier to maintain and update the data in Top-down approach.

Disadvantages

1. The Cost, time Taken to Designing and its Maintenance is very High.
2. Top-down approach is ^{not} suitable for organization that Requires High Flexibility in their Data Reporting & analysis.
3. Top-down approach can be expensive to implement and maintain for particularly small organization.

(II) Bottom-up Approach



1. First, data is extracted from external source (same as happens in Top-down approach).
2. Then, the data go through staging area (as explained in Top-down approach) and loaded to Data marts instead of Data warehouse. The Data mart ~~are~~ are created first and provide Reporting capability.
3. These data marts are then Integrated into Data warehouse.

This approach is given by Kimball as — data marts ~~are~~ are created first and provides a thin view for analyse and ~~at~~ data warehouse is created after complete Data marts have been created.

Advantages :-

1. As Data marts are created first, so the Reports are quickly generated.
2. Also, the cost and Time Taken in Designing this model is low comparatively.
3. The Bottom-up approach is more flexible than Top-down approach.
4. The Bottom-up approach Reduces Risk of failure, as Data mart ^{can be} Tested and Refined before Being incorporated into Large data warehouse.

Disadvantages :-

1. This is not strong as Top-down approach as Dimensional view of Data mart is not consistent.
2. The Bottom-up approach is more complex than Top-down approach.